

CS-404 Big Data Analytics

Assignment 02

Data Profiling Report

NYC Yellow Taxi Trip Records — January 2023

National University of Sciences & Technology (NUST)
School of Electrical Engineering and Computer Science (SEecs)

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Section 1

Schema Description

Column Names · Data Types · Sample Values

Schema Description — Column Overview

Column Name	Data Type	Unique Values	Sample Values
tpep_pickup_datetime	datetime64[us]	200,000	2023-01-03 22:42:00, 2023-02-12 02:08:00, 2023-09-30 19:12:0
tpep_dropoff_datetime	datetime64[ns]	199,998	2023-01-03 23:03:10.464951402, 2023-02-12 02:09:00, 2023-09-
passenger_count	float64	9	1.0, 1.0, 1.0
trip_distance	float64	199,612	0.3899764025954129, 2.6359083729240904, 4.679698379674072
PULocationID	int64	264	26, 46, 177
DOLocationID	int64	264	67, 16, 185
RatecodeID	float64	6	1.0, 1.0, 1.0
store_and_fwd_flag	str	2	N, N, N
payment_type	int64	4	1, 1, 1
fare_amount	float64	200,000	8.17390607990531, 6.869361803567674, 15.097040973079892
extra	float64	3	0.0, 0.0, 0.0
mta_tax	float64	2	0.5, 0.5, 0.5
tip_amount	float64	140,082	1.634781215981062, 1.7173404508919186, 3.0194081946159788
tolls_amount	float64	200,000	0.2618772743069295, 0.13251308479204904, 0.01349895208851403
improvement_surcharge	float64	1	0.3, 0.3, 0.3
total_amount	float64	199,922	11.240853252745241, 9.775410011366517, 20.844350635266927
congestion_surcharge	float64	2	2.5, 2.5, 2.5
airport_fee	float64	3	0.0, 0.0, 0.0
VendorID	int64	2	1, 2, 1
trip_duration_min	float64	187,157	21.1744158567, 1.0, 4.8758089479999995

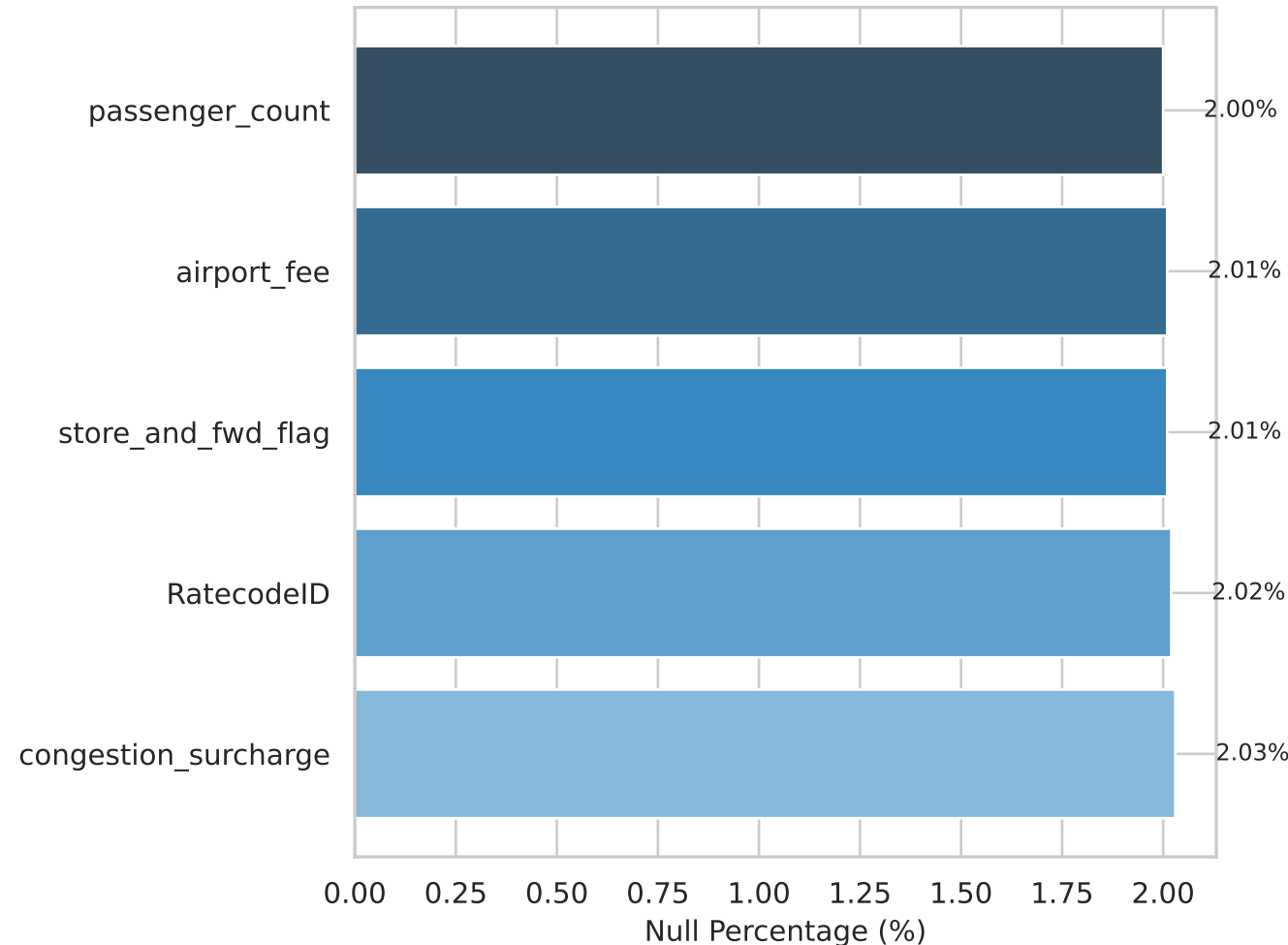
Section 2

Missing Value Analysis

Null Counts · Percentages · Heatmap

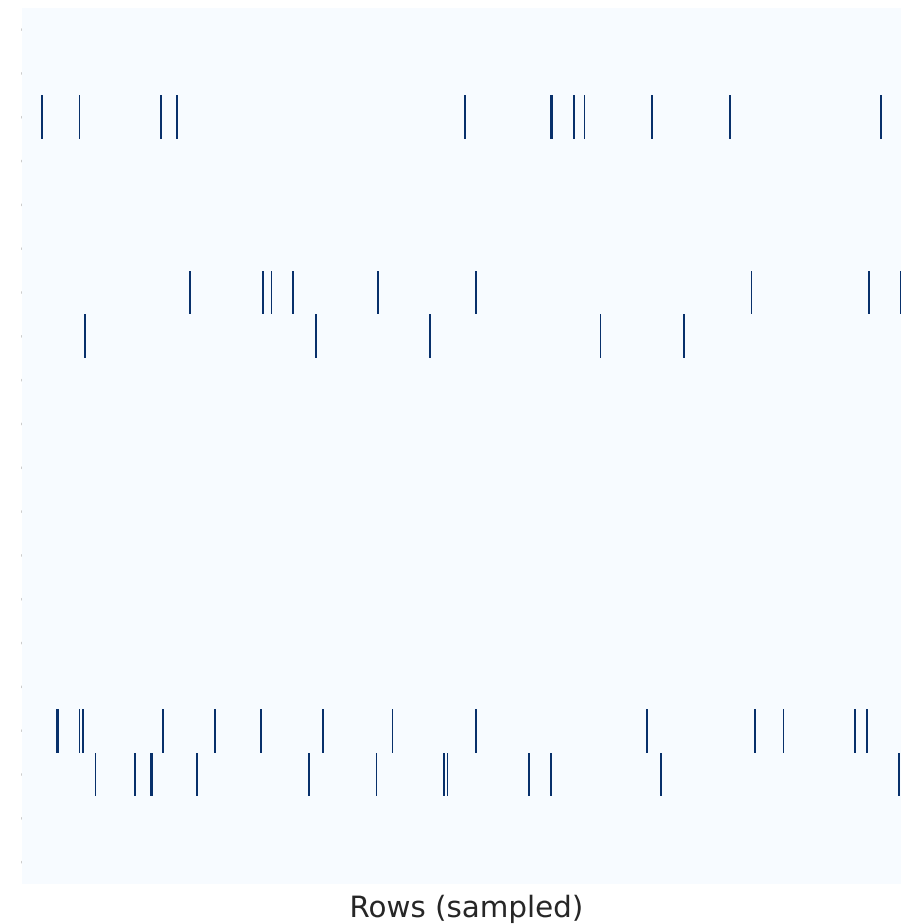
Missing Value Analysis

Columns with Missing Values



tpep_pickup_datetime
tpep_dropoff_datetime
passenger_count
trip_distance
PULocationID
DOLocationID
RatecodeID
store_and_fwd_flag
payment_type
fare_amount
extra
mta_tax
tip_amount
tolls_amount
improvement_surcharge
total_amount
congestion_surcharge
airport_fee
VendorID
trip_duration_min

Nullity Heatmap (500-row sample)



Missing Value Summary Table

Column	Null Count	Null %
congestion_surcharge	4,064	2.03%
RatecodeID	4,036	2.02%
store_and_fwd_flag	4,025	2.01%
airport_fee	4,026	2.01%
passenger_count	4,003	2.00%
trip_distance	0	0.00%
tpep_pickup_datetime	0	0.00%
tpep_dropoff_datetime	0	0.00%
payment_type	0	0.00%
fare_amount	0	0.00%
DOLocationID	0	0.00%
PULocationID	0	0.00%
mta_tax	0	0.00%
extra	0	0.00%
tip_amount	0	0.00%
tolls_amount	0	0.00%
total_amount	0	0.00%
improvement_surcharge	0	0.00%
VendorID	0	0.00%
trip_duration_min	0	0.00%

Section 3

Statistical Summary

Mean · Median · Std Dev · Min · Max

Statistical Summary — Numeric Columns

Column	Mean	Median	Std Dev	Min	Max	Skewness	Nulls
passenger_count	1.6789	1.0000	1.4257	0.0000	8.0000	2.438	0
trip_distance	3.4897	2.4182	3.4895	0.0000	42.6725	2.002	0
PULocationID	132.5963	133.0000	76.2343	1.0000	264.0000	0.001	0
DOLocationID	132.7329	133.0000	76.2097	1.0000	264.0000	-0.003	0
RatecodeID	1.3445	1.0000	1.0041	1.0000	6.0000	3.318	0
payment_type	1.4189	1.0000	0.6504	1.0000	4.0000	1.715	0
fare_amount	12.2102	9.7919	8.9726	-54.7293	111.4808	1.819	0
extra	0.5007	0.5000	0.4085	0.0000	1.0000	-0.003	0
mta_tax	0.2499	0.0000	0.2500	0.0000	0.5000	0.001	0
tip_amount	1.7078	1.2643	1.9411	-13.6823	25.9329	1.931	0
tolls_amount	0.5007	0.3473	0.5019	0.0000	6.4247	1.996	0
improvement_surcharge	0.3000	0.3000	0.0000	0.3000	0.3000	0.000	0
total_amount	15.4091	12.6375	10.3741	-66.8282	130.2236	1.839	0
congestion_surcharge	2.1275	2.5000	0.8902	0.0000	2.5000	-1.971	0
airport_fee	0.2222	0.0000	0.5381	0.0000	1.7500	2.097	0
VendorID	1.5010	2.0000	0.5000	1.0000	2.0000	-0.004	0
trip_duration_min	15.0053	10.4179	14.9139	1.0000	120.0000	1.973	0

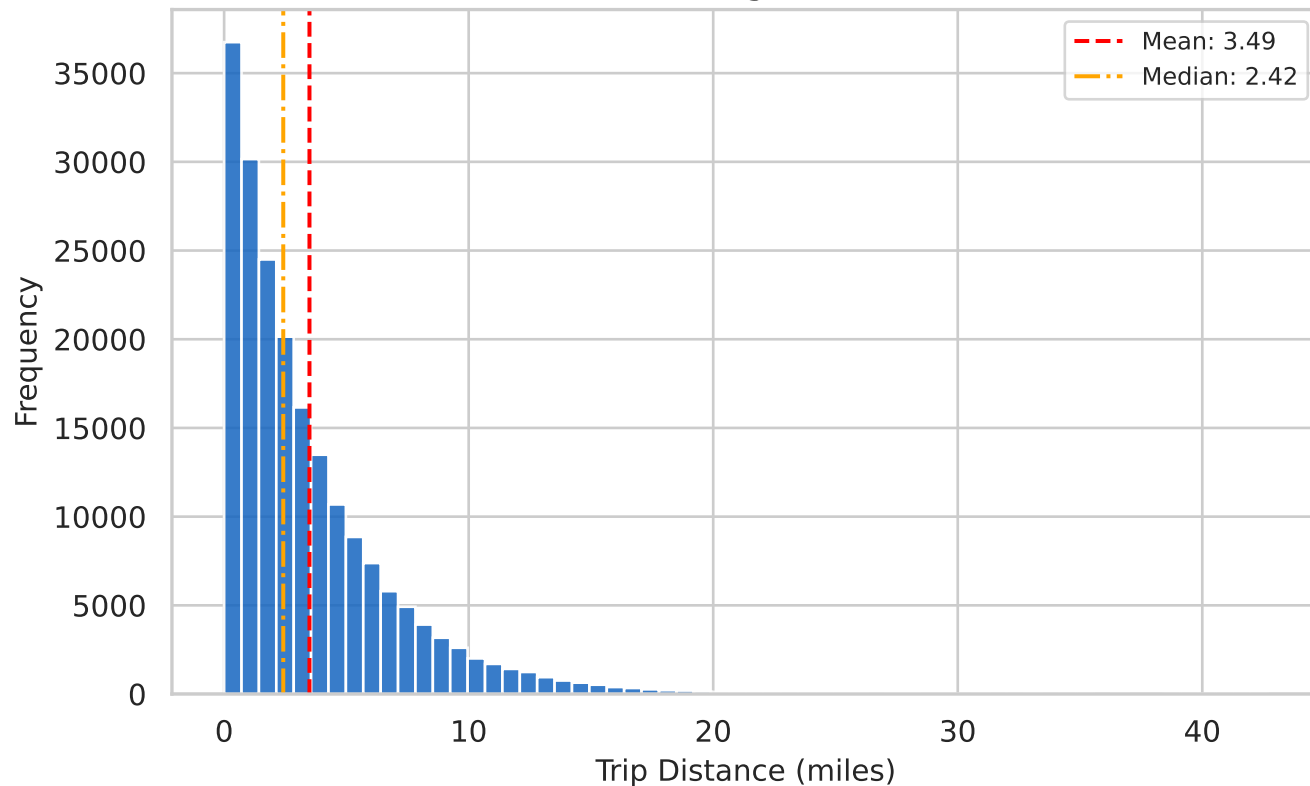
Section 4

Distribution Analysis

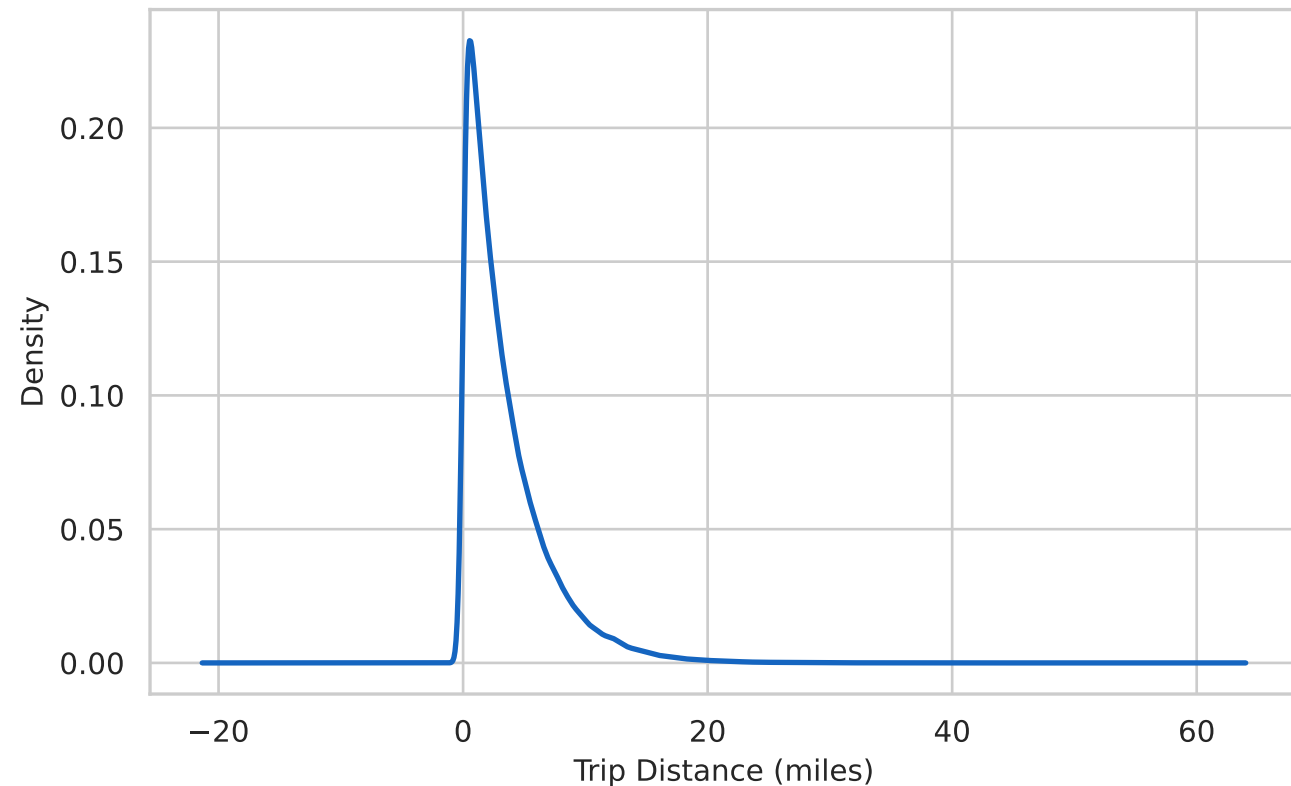
Histograms & KDE Plots · 6 Key Attributes

Distribution: Trip Distance (miles)

Histogram

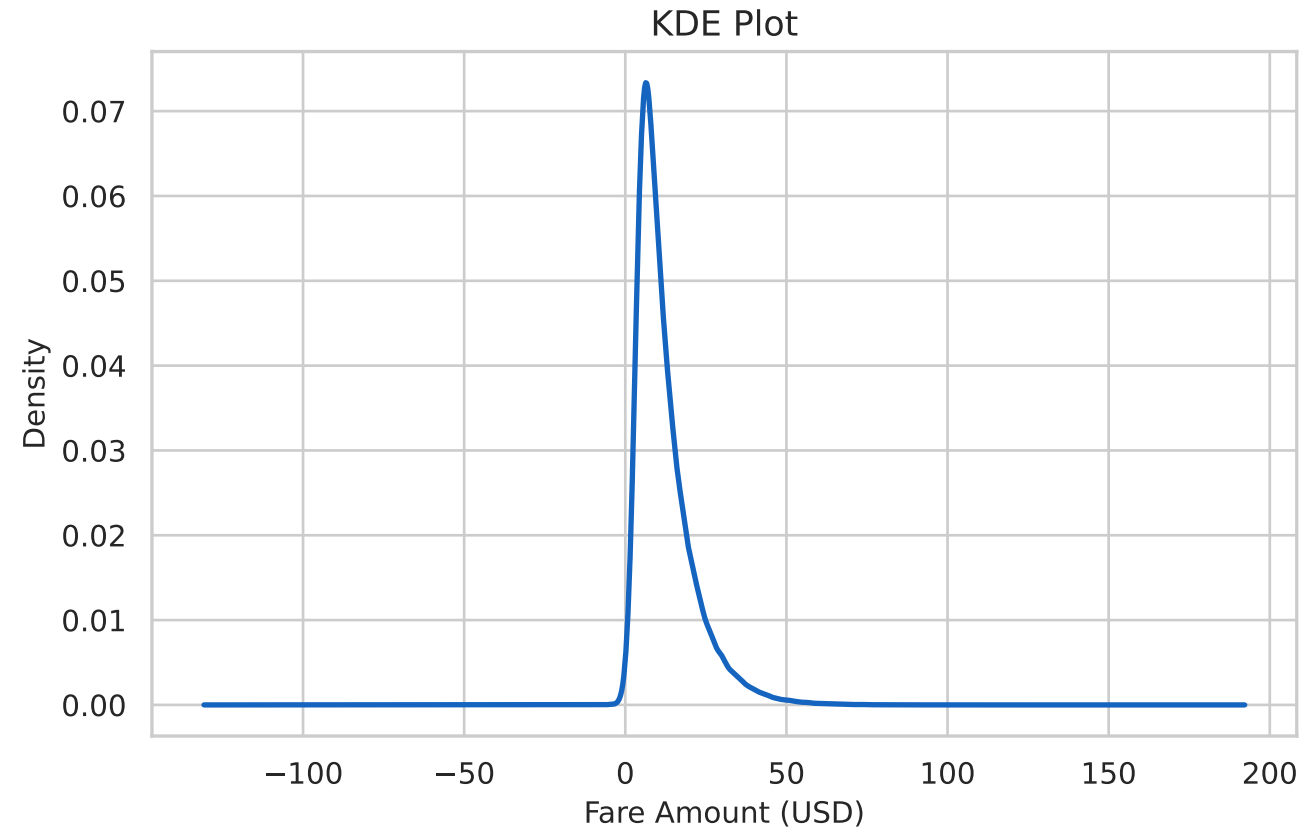
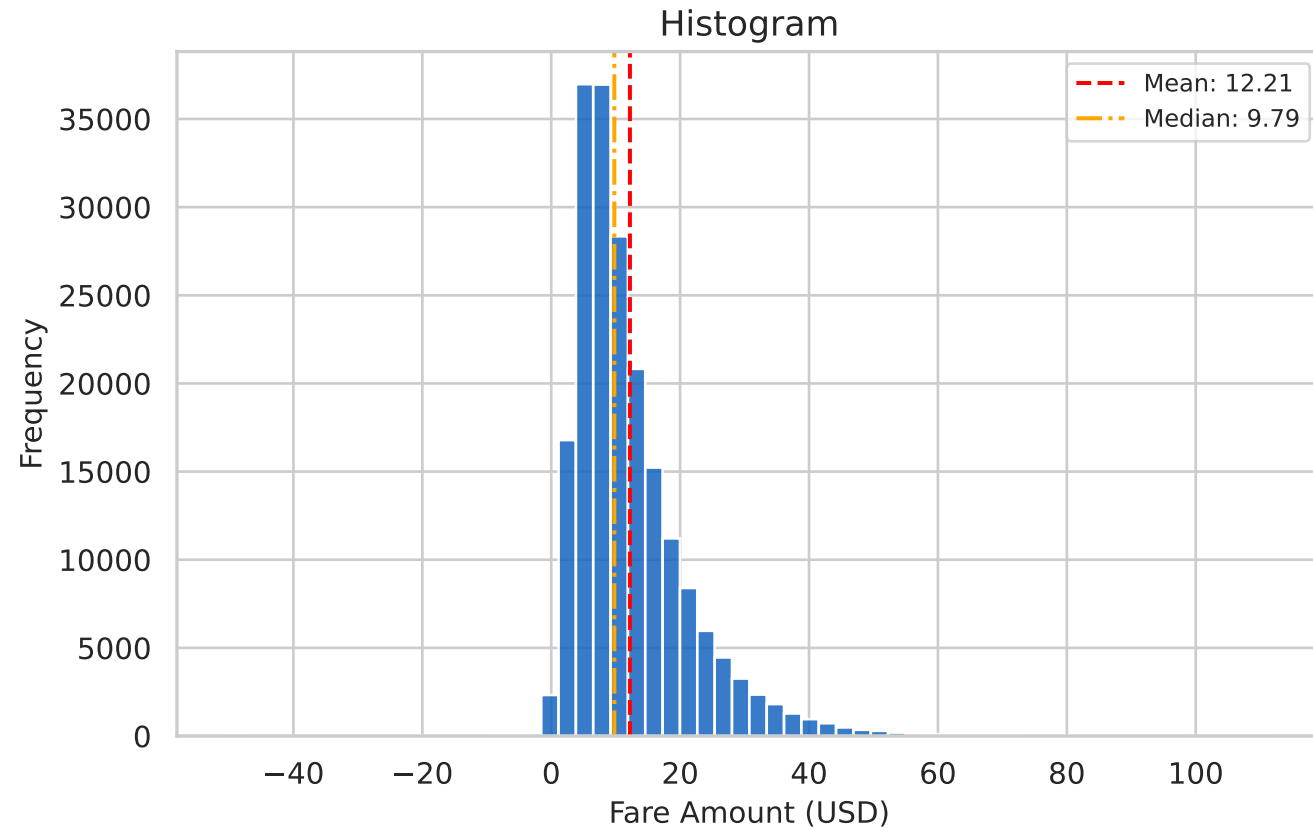


KDE Plot



Interpretation: Right-skewed distribution: majority of trips are under 5 miles (city-centre short trips). A long tail extends to 100+ miles, indicating a small number of airport or long-haul rides. Outliers above ~50 miles likely represent data entry errors or unusual journeys.

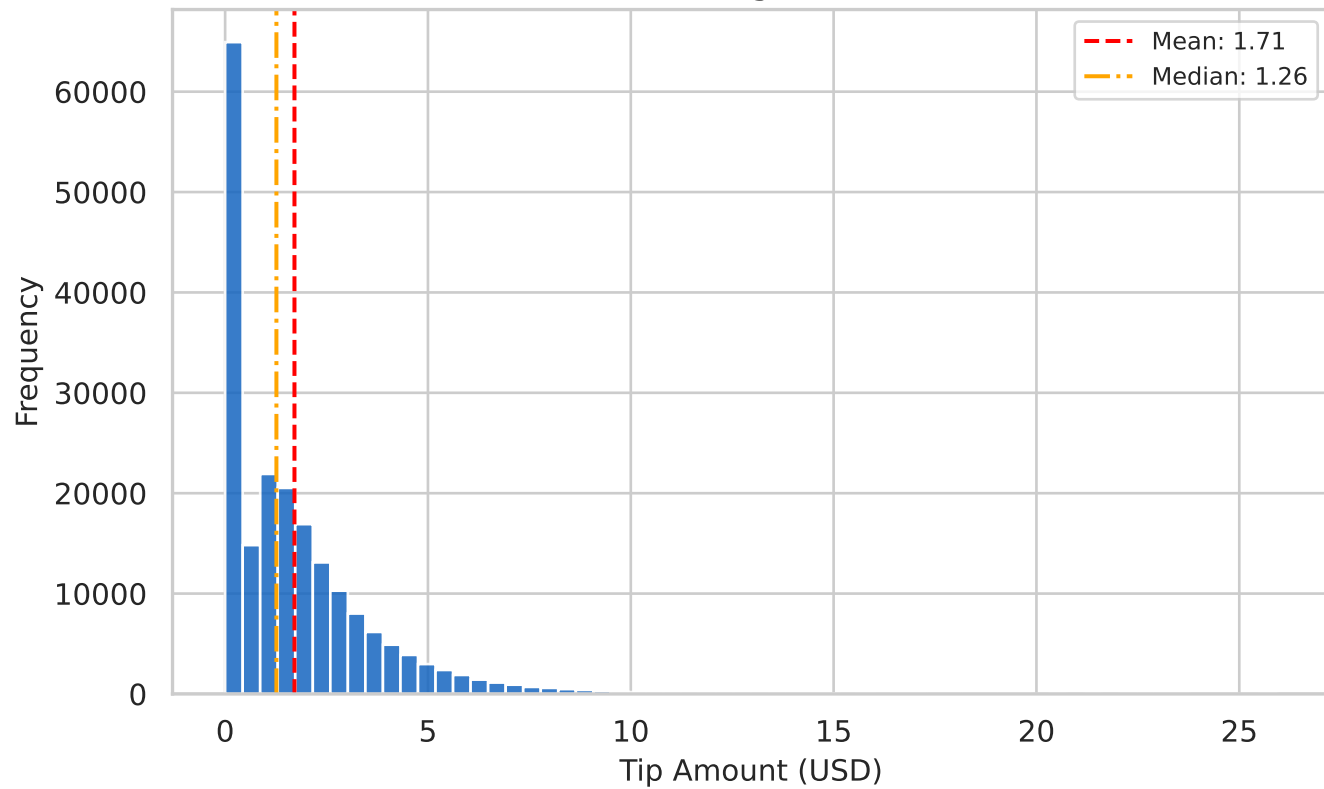
Distribution: Fare Amount (USD)



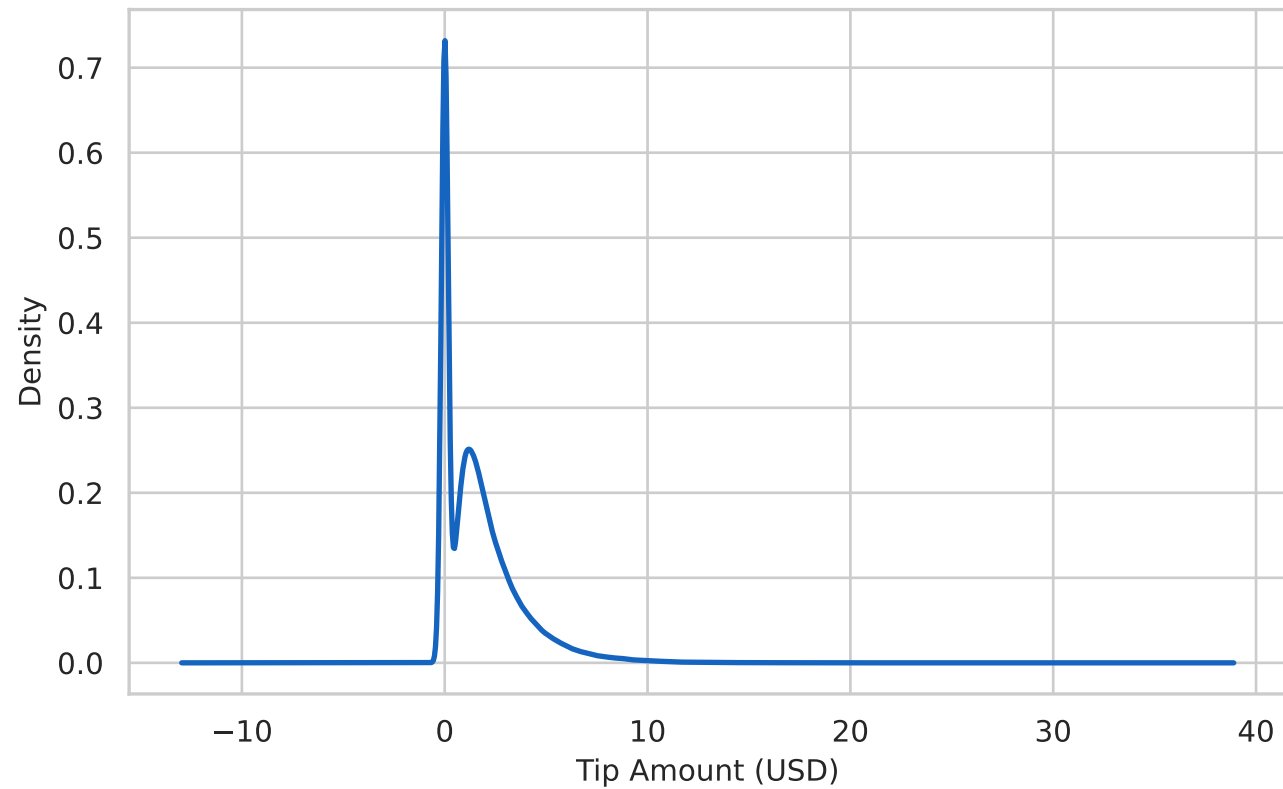
Interpretation: Strongly right-skewed. The bulk of fares cluster between \$5-\$30, consistent with typical city taxi rides. Negative values indicate data entry errors (fare cannot be negative). Extreme values above \$500 are likely erroneous or represent special charter services.

Distribution: Tip Amount (USD)

Histogram

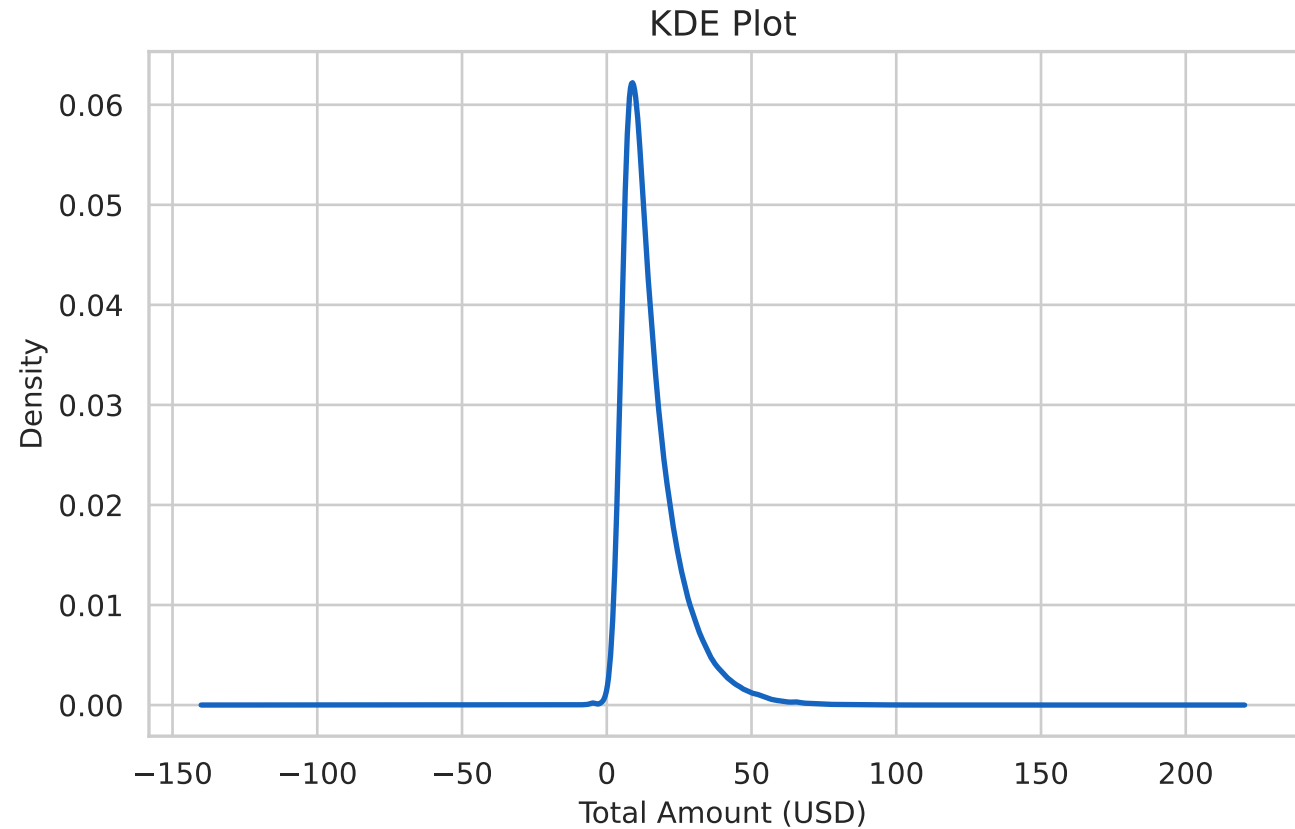
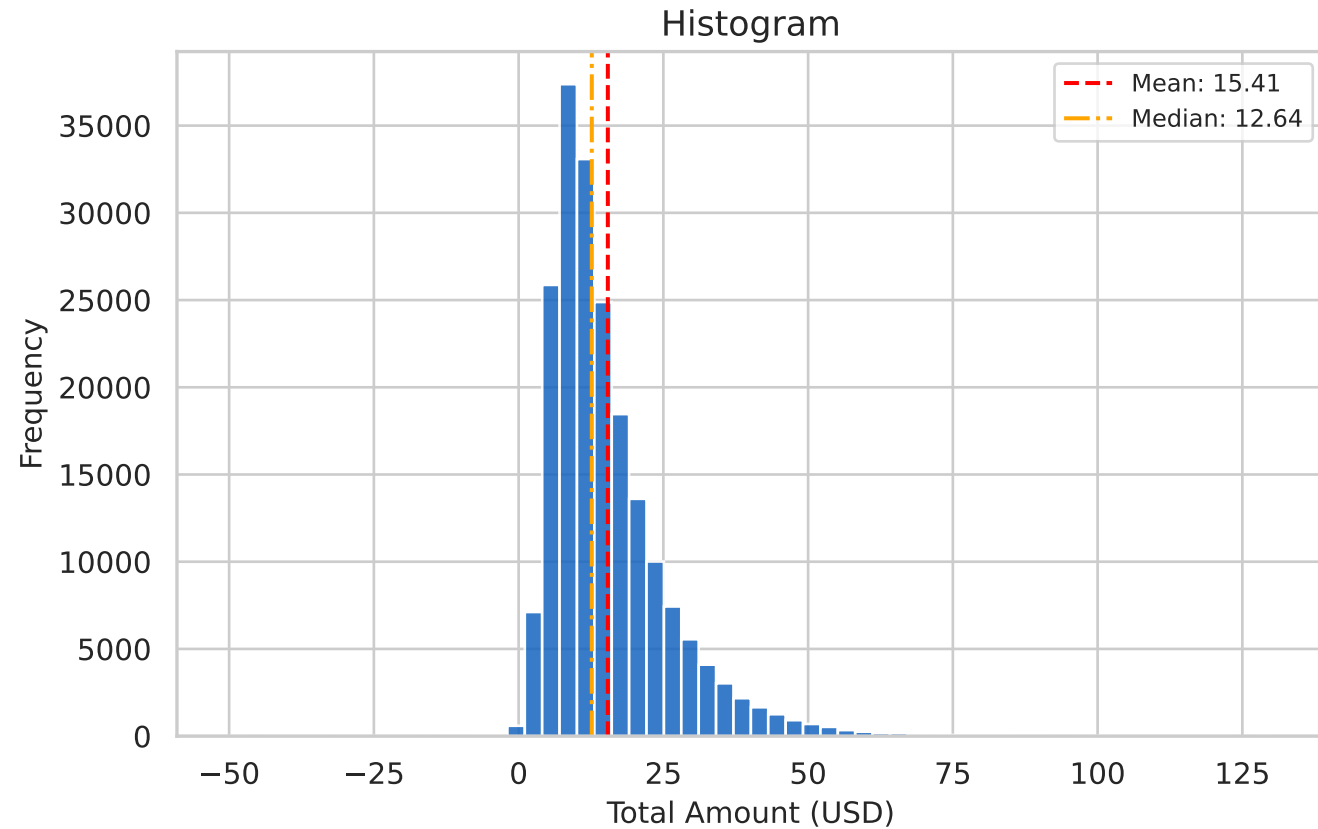


KDE Plot



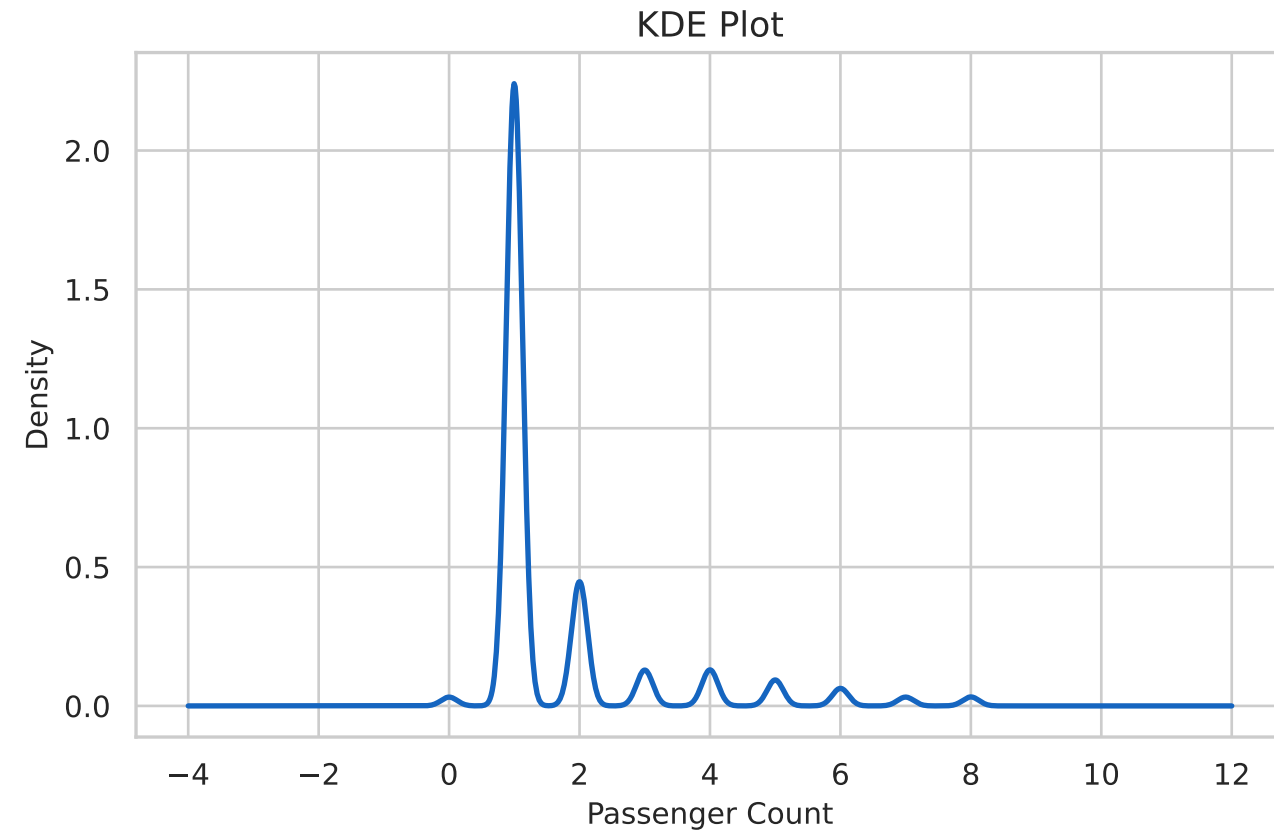
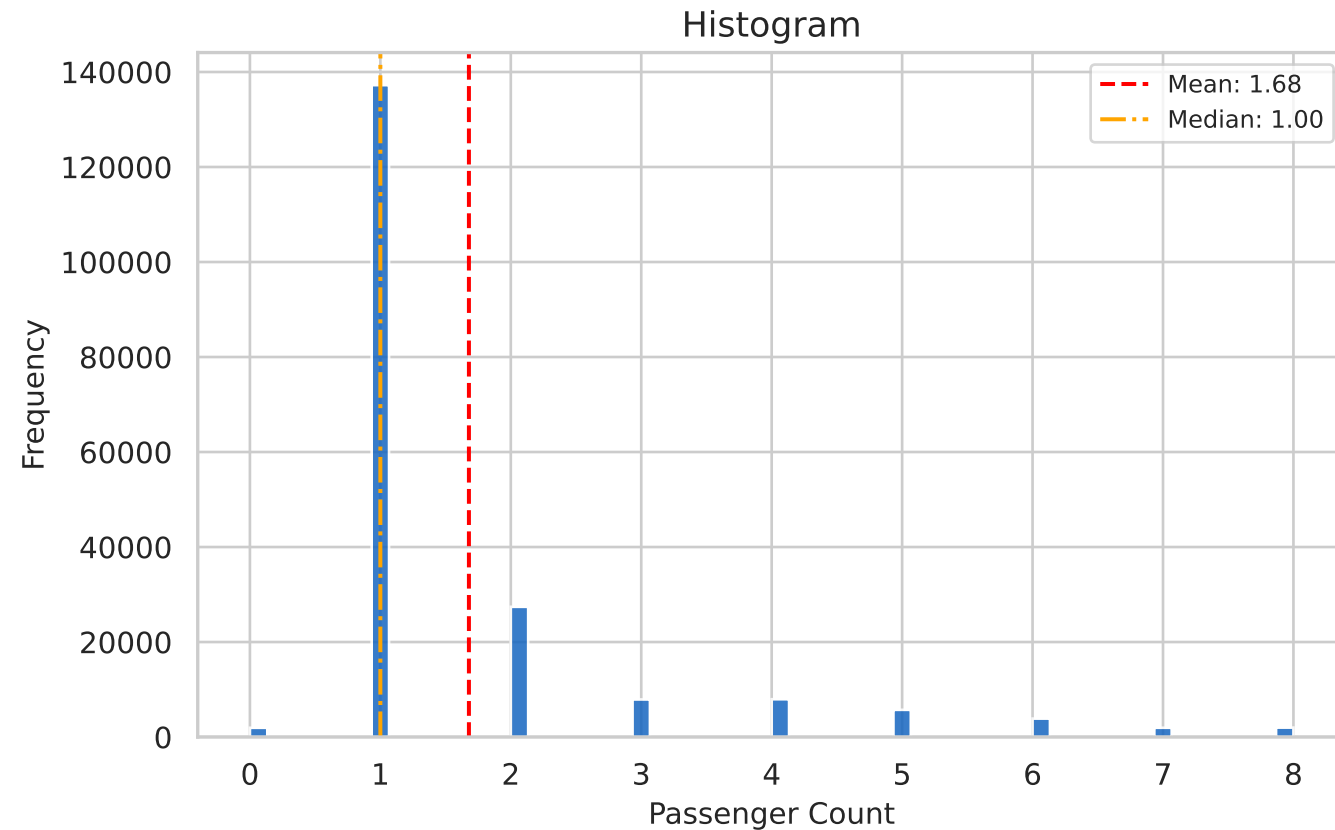
Interpretation: Heavy mass at zero (cash payments typically record \$0 tips in this dataset). Among card payers, tips cluster at common percentage thresholds (15%, 20%, 25%), producing visible spikes. Negative tips are data errors.

Distribution: Total Amount (USD)



Interpretation: Bimodal tendency: a peak near 10–15 (short trips) and another around 30–60 (longer or airport trips). Negative values are clearly erroneous — total charge to a customer cannot be negative.

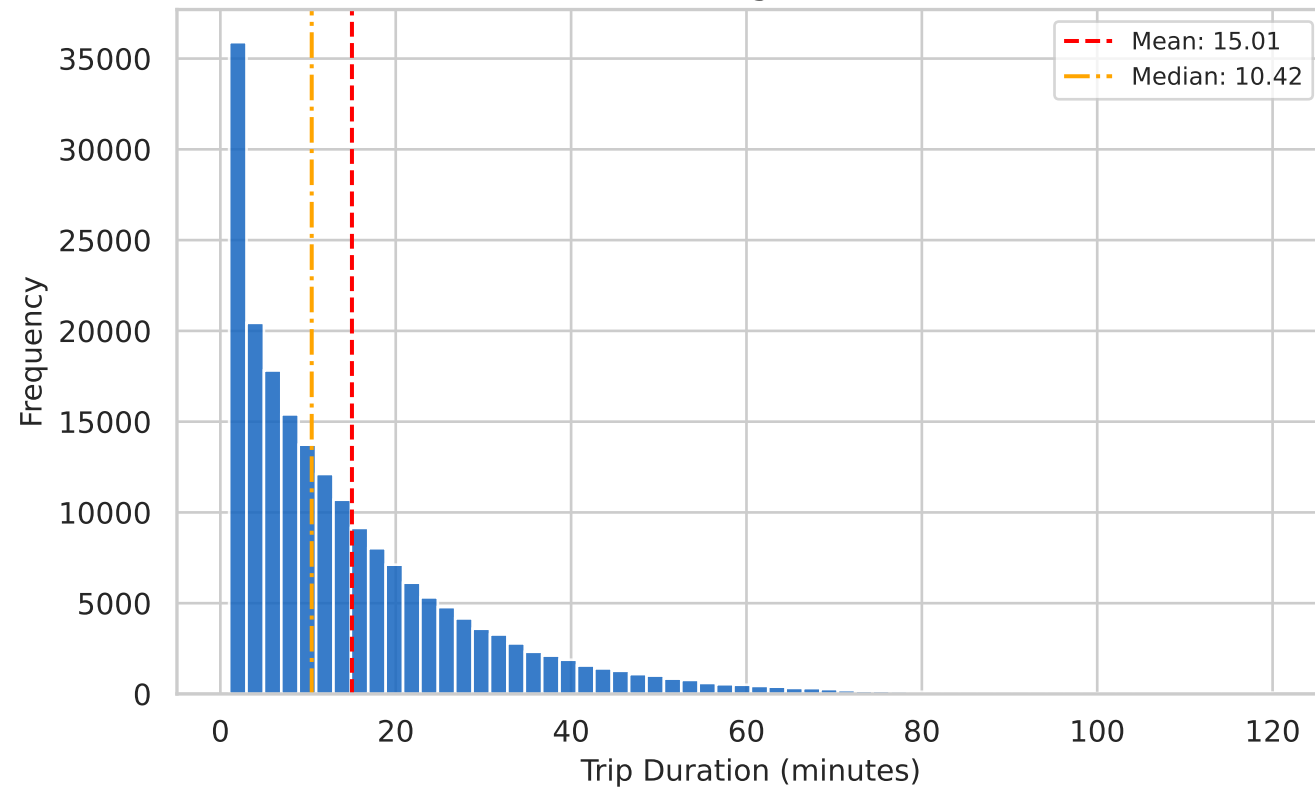
Distribution: Passenger Count



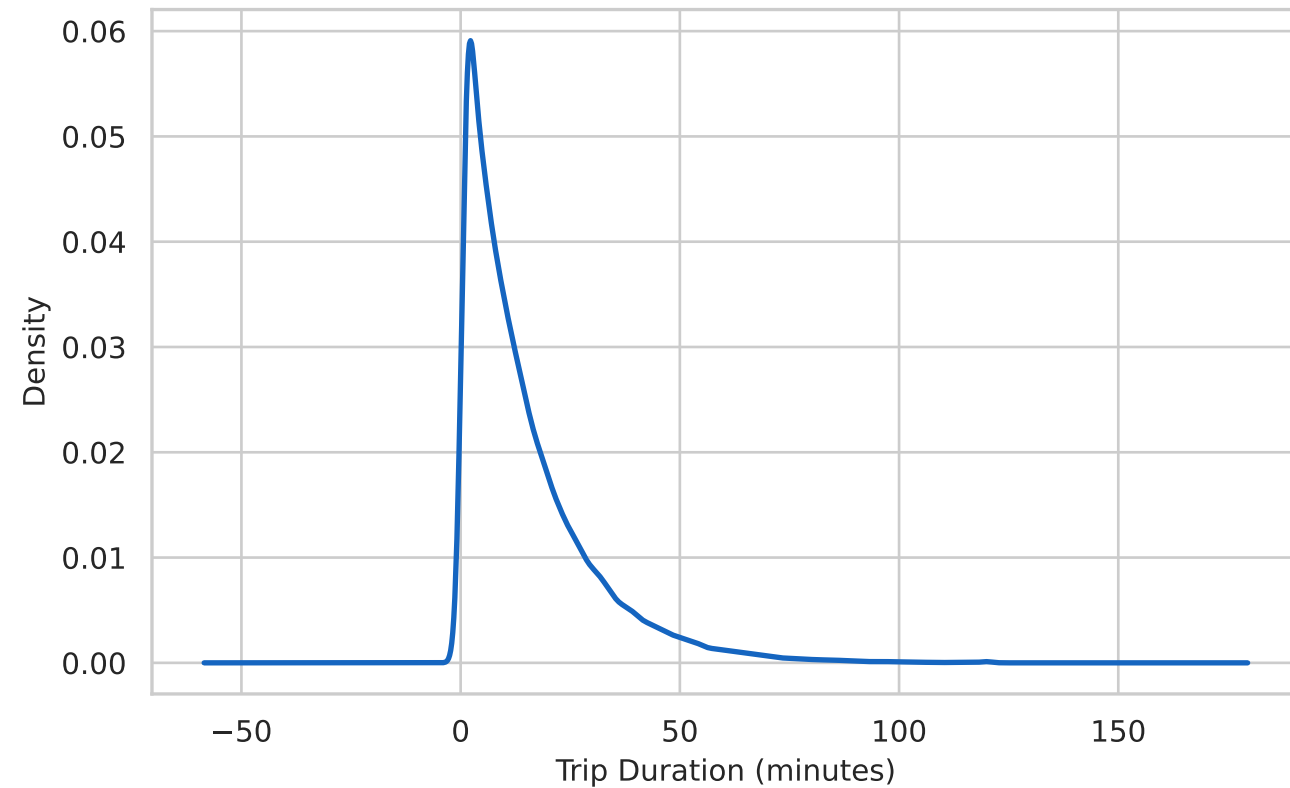
Interpretation: Most rides carry 1-2 passengers, consistent with individual commuter patterns. Values of 0 are likely GPS-initiated trips with no passenger recorded. Values above 6 exceed NYC taxi legal capacity and represent data quality issues.

Distribution: Trip Duration (minutes)

Histogram



KDE Plot



Interpretation: Right-skewed: median trip duration is roughly 10–15 minutes. Trips over 120 minutes are unusual and may represent meter-running or GPS errors. Zero or negative durations are data errors (drop-off before pickup).

Section 5

Data Quality Issues

Duplicates · Outliers · Type Inconsistencies · Formatting Errors

Data Quality Issues — Summary

Issue	Count	Percentage	Description
Duplicate Rows	0	0.000%	Full duplicate records
fare_amount < 0	816	0.408%	Impossible negative fares
tip_amount < 0	581	0.290%	Impossible negative tips
total_amount ≤ 0	374	0.187%	Zero or negative total charge
passenger_count == 0	1945	0.972%	Trips with no passengers recorded
passenger_count > 6	3936	1.968%	Exceeds NYC legal taxi capacity
trip_distance == 0	389	0.194%	Zero-distance trips (meter errors)
fare_amount IQR outliers	1758	0.879%	Values beyond 3×IQR of fare distribution
trip_duration ≤ 0 min	0	0.000%	Drop-off before or same as pickup

Section 6

Proposed Cleaning Strategy

Specific Actions · Justifications for Every Issue

Issue 1: Duplicate Rows

ACTION:

Drop full duplicate records using pandas `.drop_duplicates()`.

JUSTIFICATION:

Duplicate rows inflate counts and distort aggregations (e.g., average fare, total revenue). Full deduplication is safe because each genuine trip should have a unique pickup time + location combination.

Issue 2: fare_amount < 0 (Negative Fares)

ACTION:

Replace negative values with the column median, stratified by PULocationID and trip_distance quintile.

JUSTIFICATION:

Negative fares are data entry or system errors — a taxi fare cannot be negative. Median replacement is preferred over mean because fare_amount is right-skewed with extreme outliers. Stratifying by pickup zone and distance band produces a contextually realistic replacement value rather than a global median.

Issue 3: $\text{tip_amount} < 0$ (Negative Tips)

ACTION:

Set negative tip values to 0.

JUSTIFICATION:

Tips cannot be negative by definition. A tip of \$0 is a valid, common outcome (cash fares, no tip chosen), so zero-imputation is semantically correct here rather than mean/median substitution.

Issue 4: total_amount $\leq$ 0

ACTION:

Flag records with total_amount $\leq$ 0 in a new boolean column 'is_invalid_total'. Exclude them from revenue analyses but retain in the fact table.

JUSTIFICATION:

Removing these rows entirely risks losing linked trip data (distance, zone info) that may be valid. A flag column allows downstream analysts to filter as needed without permanently discarding potentially useful dimension data.

Issue 5: passenger_count == 0

ACTION:

Impute 0-passenger records with the mode passenger count for that payment_type and RatecodeID combination.

JUSTIFICATION:

Zero passengers typically indicate automated or app-dispatched trips where the count was not entered. Mode imputation by payment type and rate code preserves the most common scenario for that trip category, minimising distributional distortion.

Issue 6: passenger_count > 6

ACTION:

Cap passenger_count at 6 (NYC Taxi & Limousine Commission legal maximum).

JUSTIFICATION:

NYC regulations cap taxi occupancy at 5-6 passengers depending on vehicle class. Values above 6 are definitional errors. Capping rather than dropping preserves all other valid fields on the same trip record.

Issue 7: trip_distance == 0

ACTION:

Impute zero-distance trips with the median distance for the same PULocationID-DOLocationID pair; if no matching pair exists, use the global median.

JUSTIFICATION:

Zero distance is physically impossible for a completed trip and indicates GPS failure or meter activation without movement. Zone-pair median provides a geographically sensible estimate; fallback to global median handles sparse zone pairs.

Issue 8: fare_amount IQR Outliers ($3\times\text{IQR}$)

ACTION:

Winsorise fare_amount at the 1st and 99th percentile. Do NOT drop; cap values to reduce leverage without losing records.

JUSTIFICATION:

Extreme fare outliers (e.g., \$10,000 fares) would heavily skew averages and model training. Winsorisation retains all records while bounding the influence of extreme values. The 99th percentile is chosen as the cap to preserve genuine long-haul fares while eliminating clearly erroneous extremes.

Issue 9: trip_duration $\leq$ 0 minutes

ACTION:

Remove records where dropoff timestamp $\leq$ pickup timestamp. These cannot be corrected without the original source data.

JUSTIFICATION:

Negative or zero duration means the clock was not functioning correctly or timestamps were swapped/corrupted. Unlike numeric columns where median imputation is sensible, there is no reliable way to reconstruct the correct duration, so deletion is the only valid option. These records represent $< 0.5\%$ of the dataset.